Machine Learning Laboratory
Lab Assignment: Simple Linear Regression
Lab 3

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Simple Linear Regression -
Simple Linear Regression is a supervised learning method used to study the relationship between one input variable and one continuous output variable. It fits a straight line to the data and uses this line to make predictions.

In this experiment, vehicle weight is used as the input variable and MPG (Miles Per Gallon) is used as the target variable. The main purpose is to understand Linear Regression mathematically and implement its important parts from scratch before comparing the results with Scikit-learn.

Dataset Description -
The Auto MPG Dataset was used for this experiment. The dataset contains information about different vehicles and their fuel efficiency.
For this experiment, only two variables were selected:

- Input (X): weight
- Target (y): mpg

The complete dataset contains 398 observations. The dataset was divided into 80% training data and 20% testing data using the same split for both the manual and Scikit-learn models.

Methodology -

Load and inspect the dataset.
- Check missing and duplicate values.
- Select weight and MPG.
- Perform an 80:20 train-test split.
- Implement prediction manually.
- Calculate prediction error and squared-error loss.
- Implement the MSE cost function.
- Calculate slope and intercept manually.
- Generate predictions using the manual model.
- Calculate MAE, MSE, RMSE and R^2 manually.
- Train the same model using Scikit-learn.
- Compare both implementations.
- Perform residual analysis.

The assignment requires the major prediction, loss, cost, parameter and evaluation calculations to be implemented manually before using Scikit-learn.

Workflow Flowchart

Load Auto MPG Dataset -> Explore Dataset -> Check Missing Values & Duplicates -> Select Weight → MPG -> 80% Train / 20% Test -> Calculate Mean -> Calculate Slope & Intercept -> Generate Predictions -> Calculate Errors & Loss -> Calculate MAE, MSE, RMSE, R^2 -> Scikit-learn Model -> Compare Results -> Residual Analysis -> Final Conclusion

Mathematical Model

Simple Linear Regression equation is: $\hat{y}=\beta_0+\beta_1 x$ where:

- β_0 = intercept
- β_1 = slope
- x = vehicle weight
- $\hat{y}$ = predicted MPG

prediction error - $e_i=y_i-\hat{y}_i$
squared error – $L_i=(y_i-\hat{y}_i)2$
Mean Squared Error – $1/n \sum n(y_i-\hat{y}_i)2$

Manual Calculation of Model Parameters :

Parameter	Value
Mean Vehicle Weight	2969.0157
Mean MPG	23.6082
Slope	-0.00780524
Intercept	46.7821

fitted regression equation is: MPG = 46.7821− 0.00780524(Weight)
note - The negative slope shows that heavier vehicles generally have lower MPG.

Model Evaluation : The manually calculated MSE costs :

Dataset	MSE
Training	19.7819
Testing	14.8949

The testing MSE is reasonably close to the training MSE, indicating that the model gives a similar level of error on the testing data.

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MAE: 3.117786199206457 MSE: 14.8948610646362 RMSE: 3.859386099451077 R²:
0.7229710573030748
```

Residual Analysis

Residuals were calculated using:

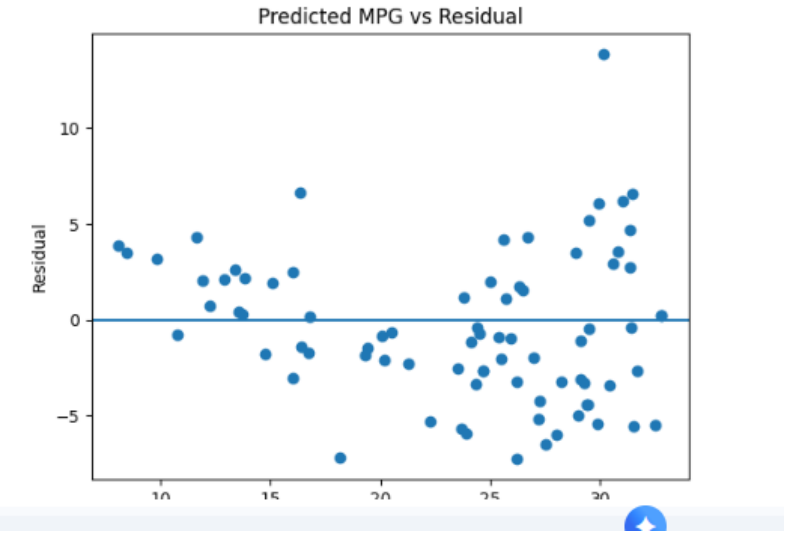
$e_i = y_i - \hat{y}_i$

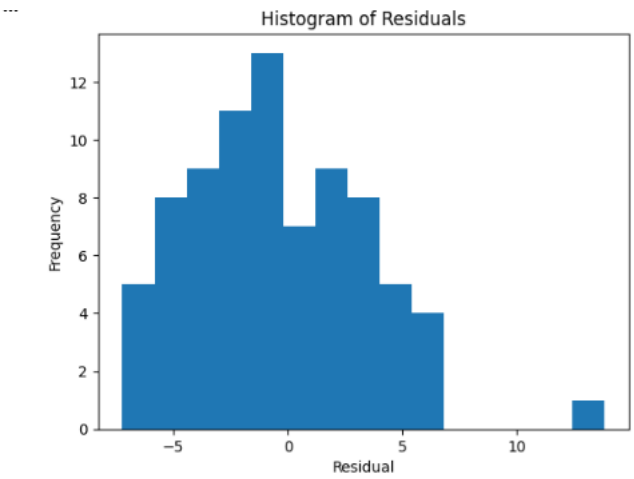
Two plots were generated:

1. Predicted MPG vs Residual
2. Histogram of Residuals

mean = - 0.410966997045316

The residual plot was examined to check whether the residuals were randomly distributed around zero. A random distribution around zero generally indicates that the linear model has captured the main relationship in the data.





Ananlysis -

The experiment shows a clear **negative relationship between vehicle weight and MPG**. As vehicle weight increases, the predicted MPG decreases. The fitted line therefore has a negative slope.

The model provides a reasonable prediction using only vehicle weight, but it cannot explain every difference in MPG. Some vehicles with similar weights can have different MPG values because other vehicle characteristics also affect fuel efficiency.

The residual analysis is useful for identifying errors and possible patterns that are not captured by the linear model. Therefore, although weight is an important predictor, using only one feature limits the overall prediction ability of the model.

Task 1

Q1**.** No. A strong relationship does not mean weight alone can predict MPG accurately because other factors also affect MPG.

Q2**.** It shows that weight alone cannot explain all differences in MPG.

Q3. No. Correlation does not by itself prove that increasing weight directly causes MPG to decrease.

Task 2

Q1. Test data should not be used because it would cause information from the test set to influence the model.

Q2. Duplicates can make the model appear better because similar observations may occur in both training and testing data.

Q3. A different split gives different training and testing samples, so the fitted model and R^2 can change.

Task 4

Q1. Squaring makes both +5 and -5 equal to 25.

Q2. Squaring gives much more importance to large errors.

Q3. Positive and negative errors can cancel each other, making the total error misleading.

Task 5

Q1. A low average MSE does not guarantee that every prediction is reliable because a few large errors may still exist.

Q2. MSE squares errors, so it is more affected by large errors than MAE.

Q3. The MSE becomes one-fourth of its original value.

Task 6

Q1. Deviations from the mean show how the input and target values vary around their averages.

Q2. The denominator becomes zero, so the slope cannot be calculated.

Q3. The relationship remains the same, but the numerical slope changes because the unit of weight changes.

Task 7

Q1. The model uses only weight, so the same weight produces the same prediction.

Q2. Yes. A regression equation can produce a value that was not exactly present in the training data.

Q3. It is extrapolation, so the model has less information about that range and the prediction can be less reliable.

Task 8

Q1. The line minimizes the overall error, so it normally cannot pass through every observation.

Q2. It represents the residual or prediction error.

Q3. The dataset with points closer to the line would generally be more reliable because its prediction errors are smaller.

Task 9

Q1. No. A high R^2 does not guarantee that there are no large individual errors.

Q2. RMSE gives more weight to large errors, while MAE treats errors more evenly.

Q3. A negative R^2 means the model performs worse than simply predicting the mean MPG.

Task 10

Q1. I would first check the manual slope, intercept and training data.

Q2. Both methods use the same data and the same Linear Regression relationship, so their results should be nearly identical.

Q3. Manual implementation helps demonstrate an understanding of the mathematical working behind the algorithm.

Task 11

Q1. I would check the prediction function and the input values used for prediction.

Q2. No. The same R^2 can occur even when individual predictions are different.

Q3. Comparing coefficients, predictions and evaluation metrics together gives stronger evidence that the implementation is correct.

Task 12

Q1. A curved residual pattern suggests that a simple linear relationship may not fully describe the data.

Q2. It indicates that the regression line is systematically overpredicting or underpredicting in different weight ranges.

Q3. Random residuals around zero suggest that there is no obvious pattern left for the model to capture.

Task 13

Q1. Other vehicle characteristics affect MPG, so weight cannot explain every observation.

Q2. Horsepower could be useful because engine power can affect fuel consumption.

Q3. No. A strong residual pattern suggests that the model is missing some important relationship even if its R^2 is reasonably high.

11. Conclusion

Simple Linear Regression was successfully implemented using vehicle weight to predict MPG. The manually calculated slope was negative, showing that heavier vehicles generally have lower fuel efficiency.

The model was evaluated using different error measures and was also compared with the Scikit-learn implementation. The manual and library approaches produced nearly identical regression results when the same data was used.

The experiment also showed the limitation of using only one predictor. Weight is useful for predicting MPG, but additional vehicle features would likely improve the model.

12. References

1. UCI Machine Learning Repository, Auto MPG Dataset**.**
2. Kaggle, Auto MPG Dataset.
3. Scikit-learn, Linear Regression documentation.